

Practical 3: Simulation of Single Mote using TOSSIM

To simulate a single sensor node in TOSSIM and observe its execution logs, you will need a basic TinyOS application (like Blink), a Python simulation script, and a configuration file for the radio topology.

CODE:

```
T110 SANJANA SAHU.py - C:/Users/IT-02/Desktop/T110 SANJANA SAHU/T110 SANJANA SAHU.py (3.14.6)
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import heapq

class MockNode:
    """
    Represents a single simulated sensor node.
    """

    def __init__(self, node_id):
        self.node_id = node_id
        self.boot_time = None

    def boot_at_time(self, time):
        self.boot_time = time

class MockTossim:
    """
    Mock version of the TOSSIM simulator.

    It maintains an event queue containing:
    (simulation_time, event_description)

    The simulator jumps directly from one event to the next,
    just like the real TOSSIM.
    """

    TIMER_PERIOD = 512 # Timer interval in simulation ticks

    def __init__(self):
        self.event_queue = []
        self.nodes = {}

    def get_node(self, node_id):
        node = MockNode(node_id)
        self.nodes[node_id] = node
        return node

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```

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        self.nodes[node_id] = node
        return node

    def schedule_boot(self, node):
        # Schedule boot event
        heapq.heappush(
            self.event_queue,
            (
                node.boot_time,
                f"Booting node {node.node_id} at time {node.boot_time}."
            )
        )

        # Schedule first timer event
        first_fire = node.boot_time + self.TIMER_PERIOD

        heapq.heappush(
            self.event_queue,
            (
                first_fire,
                f"BlinkC: Timer 0 fired @ {first_fire}."
            )
        )

    def run_next_event(self):
        if not self.event_queue:
            return False

        time, description = heapq.heappop(self.event_queue)

        print(f"DEBUG ({self._active_node_id}): {description}")

        # If timer fired, schedule the next timer event
        if "Timer 0 fired" in description:
            next_fire = time + self.TIMER_PERIOD

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OUTPUT:

T110 SANJANA SAHU
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```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on wi
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:/Users/IT-02/Desktop/T110 SANJANA SAHU/T110 SANJANA SAHU.py ====
T110 SANJANA SAHU
=====
      TOSSIM Single Node Simulation
=====

Simulation Started...

DEBUG (0): Booting node 0 at time 10001.
DEBUG (0): BlinkC: Timer 0 fired @ 10513.
DEBUG (0): BlinkC: Timer 0 fired @ 11025.
DEBUG (0): BlinkC: Timer 0 fired @ 11537.
DEBUG (0): BlinkC: Timer 0 fired @ 12049.
DEBUG (0): BlinkC: Timer 0 fired @ 12561.
DEBUG (0): BlinkC: Timer 0 fired @ 13073.
DEBUG (0): BlinkC: Timer 0 fired @ 13585.
DEBUG (0): BlinkC: Timer 0 fired @ 14097.
DEBUG (0): BlinkC: Timer 0 fired @ 14609.

Simulation Finished.
>>>
```